

Designing Effective Input

CSE 4407

Md. Bakhtiar Hasan

Assistant Professor

Department of Computer Science and Engineering
Islamic University of Technology

August 14, 2025



Good Form Design

- What

- Preprinted papers requiring standardized responses
- Elicit and capture required information
- Often serve as source documents for system input

- Why

- Steers the course course of work
- Poor design leads to errors, frustration, wasted time, and cost
- Good design ensures data quality → GIGO principle
- Eliminate unnecessary forms

- Guidelines

- Make forms easy to fill in
- Ensure forms meet their intended purpose
- Design forms to ensure accurate completion
- Keep forms attractive

Guideline 1: Make Forms Easy to Fill In

- Goal
 - Reduce errors
 - Speed completion
 - Facilitate data entry
- Cost Factor
 - Employee time > Form printing cost
 - Optimize for user time!
- Electronic Submission
 - Can eliminate transcription
 - User keys data directly
- Key Techniques
 - Proper Form Flow
 - Logical Grouping
 - Clear Captioning

Ease of Use: Form Flow

- Principle
 - Minimize time and effort
 - Logical grouping makes forms easier to understand and complete
- **Standard Flow:** Left-to-right vs. Top-to-bottom
- Avoid illogical jumps → Causes frustration and errors
- **The Seven Sections**
 - Heading
 - Identification and Access
 - Instructions
 - Body
 - Signature and verification
 - Totals
 - Comments

Demo: [Bakerloo Expense Voucher Example](#)

Ease of Use: Captioning

- **What:** Labels that tell the user what information goes into each blank/field/box
- **Goal:** Users should instantly know what's expected → Clarity!
- **Common Types**
 - Line Captions (Above, Below, Left)
 - Boxed Captions
 - Check-off Captions (Vertical, Horizontal)
 - Table Captions (Column Headers)
- **Consistency:** Avoid mixing styles

Guideline 2: Meet the Intended Purpose

- **Question:** Does this form serve a necessary business function?
- **Avoid**
 - Collecting unnecessary data
 - “Nice to have” vs. “Need to have”
- **Specialty Forms:** Used when:
 - Different info needed by different departments/users from the same event
 - Specific physical constructions needed
 - Multi-part forms
 - Continuous feed for printers
 - Perforated stubs

Guideline 3: Ensure Accurate Completion

- **Impact:** Good design drastically reduces data collection errors
- **Goal:** Make it easy to do the right thing, hard to do the wrong thing
- **Example:** Handheld Devices/Scanners
 - Used by field workers (meter readers, inventory takers)
 - Data keyed/scanned directly at the source
 - Wireless upload or docking → Eliminates manual transcription errors
- **Focus:** Design elements within the form itself help prevent common mistakes

Guideline 4: Keep Forms Attractive

- **Importance:** Appealing forms encourage completion and reduce intimidation
- **Achieved Through**
 - **Uncluttered Look**
 - Good use of white space
 - Avoid cramming too much in
 - **Logical Flow and Layout**
 - Follows conventions
 - Uses sections effectively
 - **Visual Hierarchy:** Using different fonts (sparingly!), font sizes, bolding, and line weights to guide the eye and separate sections
- **Goal:** Make the user feel confident and comfortable using the form

Controlling Business Forms

- **Importance:** Managing forms as organizational assets
- **Responsibility:** Often a dedicated 'Forms Specialist' or the Systems Analyst
- **Key Duties**
 - **Validate Purpose:** Ensure each form is necessary and supports tasks
 - **Prevent Duplication:** Avoid multiple forms collecting the same info
 - **Design Effectiveness:** Apply good design principles
 - **Economical Reproduction:** Choose cost-effective printing/distribution
 - **Manage Availability:** Ensure forms are accessible when needed
- **Essential Practice:** Assign a unique **Form Number** and **Revision Date (MM/YYYY)** to every form → Aids version control and organization

Moving to Screens: Display and Web Form Design

- **Building on Principles:** Much of good form design applies here
- **Key Difference and Advantages**
 - **Cursor:** Always shows the user's current position
 - **Context-Sensitive Help:** Provide help *within* the form (e.g., rollover text, help icons) → Reduces clutter, support calls
 - **Hyperlinks:** Link to examples, definitions, or related pages directly
 - **Interactivity:** Dynamic updates, immediate feedback (more later!)
- **Goal:** Leverage the unique qualities of displays and the web

Four Guidelines for Display and Web Form Design

- **Similar Goals, Digital Context:** Aiming for Effectiveness, Accuracy, Ease of Use, Simplicity, Consistency, Attractiveness
- **Specific Display Guidelines**
 - Keep the display **simple**
 - Keep the display presentation **consistent**
 - Facilitate user **movement** among screens/pages
 - Create an **attractive** and **pleasing** display

Guideline 1: Keep the Display Simple

- **Less is More:** Show only necessary information for the current task → Aim for ~50% useful info density for occasional users
- **Three Screen Sections**
 - **Heading**
 - Top area
 - Software titles, menus, icons
 - **Body**
 - Middle area
 - Data entry, captions, instructions
 - Follows Left → Right, Top → Bottom flow
- **Other Techniques**
 - Context-sensitive help
 - Rollover buttons/elements
 - Pop-up windows
 - Context-sensitive help often accessed here
 - **Comments/Instructions**
 - Bottom area
 - Basic commands: Save, Next Page, Exit, etc.
 - Provides user security
 - Window management
 - Hyperlinks

Guideline 2: Keep the Display Consistent

- **Predictability = Usability:** Users learn faster and make fewer errors
- **Consistency Methods**
 - **Match Paper Forms:** If a screen replaces a paper form, mimic the layout where it makes sense
 - **Consistent Location:** Place common elements in the same place on every screen/page
 - **Consistent Grouping:** Group related information together logically and keep that grouping consistent across different parts of the application
 - **Clear Boundaries:** Don't let information groups visually overlap or blend confusingly

Guideline 3: Facilitate Movement

- **Goal:** Easy navigation between screens, pages, or sections
- **“Three-Clicks Rule”:** Heuristic
 - Users should ideally reach needed info within 3 clicks/keystrokes
 - BUT this is not a strict law → Don't sacrifice clarity for fewer clicks [1]
- **Movement Techniques**
 - **Web:** Hyperlinks are fundamental
 - **Scrolling:** Vertical/Horizontal movement within a page
 - **Context-Sensitive Pop-ups:** Jump to related tasks/info
 - **Onscreen Dialogs:** Step-by-step wizards, confirmations
 - **Visual Cues:** Feeling of “turning a page”

Guideline 4: Create an Attractive and Pleasing Display

- **Impact:** Appealing displays → Higher productivity, less supervision, fewer errors
- **Techniques**
 - **Uncluttered Appearance**
 - Ample white space around fields
 - Avoid jamming!
 - Use multiple windows/pages if needed
 - **Logical Flow:** Organize based on user tasks and mental models
 - **Judicious Use of GUI Elements**
 - Color, shading, 3D effects can enhance usability and appeal
 - Do NOT overdo it → Can become distracting
 - **Fonts**
 - Use different styles/sizes purposefully → Hierarchy, clarity
 - Avoid distracting or hard-to-read fonts
 - **Browser Testing**
 - Web forms may render differently
 - Test on various browsers/devices

Using Icons in Display Design

- **What:** Pictorial symbols for computer actions
- **Why**
 - Often grasped faster than words
 - Familiar interface (especially mobile)
- **Design Guidelines**
 - **Recognizable Shapes:** Use standard, easily understood icons where possible
 - **Limit Vocabulary:** ~20 distinct icons per application
 - **Consistency:** Use the same icon for the same action *everywhere*
 - **Meaningful:** Ensure the icon clearly relates to the action

Graphical User Interfaces

- **What:** How users interact with OS and applications using visual elements → Point-and-click interface
- **Key Features:** Allows direct manipulation via buttons, text boxes, etc.

Microsoft Access - [Customer]

Add Customer Order 6/3/2022 1:02 PM

Customer Number	02122		
Customer Name	Mariama Kah		
Street	2885 N. Washington		
Apartment	P.O. BOX 9200		
City	Arlington	State	WI Zip 53704
Telephone	(715) 222-1234		
Country	United States	High Volume Discount	☒
Email Address	maka25@gmail.com	First Time Purchase	☒
Current Balance	\$6,780.00	Customer Type	
Current Balance	\$320,000.00	☒ Individual ☐ Federal Government	
Payment Type	Corporate Charge ▼	☐ Corporate Customer ☐ Local or State Government	
		☐ Non-Profit Organization ☐ Educational Institution	

< > 🔍 ⌂ 🔄 📄 📧

GUI Control: Text Boxes

- **Function:** Basic field for entering text or numeric data
- **Design**
 - Must be large enough for expected input
 - Clear caption
 - **Alignment:** Often Left for text, Right for numbers
- **HTML5 Enhancements**
 - **Placeholder:** Grayed-out hint text inside the box → Vanishes on focus
 - **Specific Types:** Customize mobile keyboard → Email (symbol), Tel (number pad), URL
 - **Datalist:** Provides autocomplete suggestions as user types

GUI Control: Check Boxes and Option Buttons

- **Check Boxes:** Use for *non-exclusive* choices → Select zero, one, or multiple
 - **Appearance:** Square box, checked or empty
 - **Label:** Usually to the right → Order logically (common first)
 - **Grouping:** Use a border if many
- **Radio Buttons:** Use for *exclusive* choices → Select only one
 - **Appearance:** Round button, filled or empty → Selecting one deselects others in the group
 - **Default:** Often pre-select the most common option
 - **Alternative:** Use List Box if > 6 options
 - **Label:** Usually to the right → Order logically
 - **Grouping:** Often within an “option group” border/box

GUI Controls: List and Drop-Down List Boxes

- **List Box**

- Displays multiple options simultaneously in a box
- Allows selection of one or sometimes multiple items
- Use when you want options to be visible immediately

- **Drop-Down List Box**

- Space-saving alternative → Shows only the current selection with a down-arrow
- Clicking the arrow reveals the full list
- Best when screen real estate is limited
- Can display the most common choice initially

GUI Controls: Tabs, Sliders, Spin Buttons, Calendar

- **Tab Control Dialog Boxes**

- Organize complex settings/forms into multiple “pages” within one window
- Click tab label to switch pages
- Design
 - Create logical tabs
 - Put most used first
 - Include standard buttons

- **Sliders and Spin Buttons**

- For selecting a value within a continuous or stepped range
- Drag control along a track
- Up/Down arrows to increment/decrement value

- **Calendar Control (HTML5)**

- Standardized way to select dates/times
- Pops up a calendar interface
- Easier and less error-prone than typing dates
- Common on travel/booking sites

GUI Controls: Image Maps, Text Areas, Messages, Buttons

- **Image Maps**

- Define clickable “hotspots” within an image
- Clicking a spot sends coordinates (x, y) to the program
- **Use:** Clicking on a state in a map image

- **Text Areas**

- For entering larger blocks of multi-line text
- Larger than text boxes, often have scroll bars
- **Word Wrap:** Can be on (text flows automatically) or off (user hits Enter for new line)

- **Message Boxes**

- Pop-up windows for warnings, errors, confirmations, or simple feedback
- Should be clear and state necessary user action

- **Command Buttons**

- User clicks to trigger an action
- Text centered inside button
- Default button often visually indicated

Behind the Scenes: Form Controls and Values

- **Data Sending:** Web forms send data to the server as **Name/Value** pairs
- **Control Name**
 - Defined in the HTML code (E.g., `<input name="city">`)
 - Server script needs this name
- **Control Value:** How the value is determined depends on the control
 - **Text Box/Area:** Characters typed by user
 - **Check Box/Radio Button:** Predefined in HTML (usually not the visible label text)
 - **Drop-Down List:** Predefined for *selected* option
- **Client-Side Use:** Values can also be used by JavaScript *within* the browser for calculations/validation *before* submitting

Demo: [Cruise Ship Registration Form Example](#)

Hidden Fields

- **What:** Input fields on a web form that are invisible to the user
- **Purpose:** Store data that needs to be sent to the server, but should not be seen or edited by the user
- **Common Uses**
 - Passing data between multiple forms in a sequence
 - Storing session information
 - Retaining key fields needed for database updates
- **Structure:** `<input type="hidden" name="someName" value="someValue">`

Complex Interactions: Event-Response Charts

- **Problem:** GUI forms (especially web) can have complex interactions → Clicking button X changes field Y, which enables button Z...
- **Solution:** Event-Response Chart → Map user actions (Events) to system reactions (Responses)
- **Events**
 - User actions on specific controls (or the whole page)
 - **Examples:** Button click, value change, gaining focus, losing focus, page load, key press
- **Responses**
 - What the system/page should do
 - **Example:** Validate data, perform calculation, update field, display message, enable/disable button, navigate
- **Benefit:** Helps analyze, design, and document complex interface logic

Demo: [Azure Isle Cost Estimator Example](#) vs. [Event-Response Form Example](#)

Event Beyond Single Page and Dynamic Web Pages

- **Event Scope:** Events not limited to a single form/page
 - Can trigger navigation to other pages
 - Can dynamically change content on related controls
- **Dynamic Web Pages**
 - Pages that change content/appearance based on user actions without a full page reload
 - Uses client-side scripting (usually JavaScript)
 - **Examples:** Rollover image changes, expanding/collapsing menus, responsive layouts adapting to screen size, form fields appearing/disappearing based on choices
 - **Power**
 - Creates richer, more responsive user experience
 - Can customize forms on-the-fly

Dynamic Technique: Using Layers

- **Idea:** Displaying temporary info using stacked layers on a web page
 - Uses CSS z-index property to control stacking order
 - Base Layer = Main page
 - Other Layers initially hidden below
- **How**
 - **Trigger Event:** User action makes a hidden layer visible or generates with JavaScript
 - **Positioning:** Layer appears at predefined coordinates
 - **Removal Event:** User action hides or removes the layer
- **Benefits**
 - Avoids separate pop-up windows
 - No need for full page reload to show/hide info
 - Does not clutter the main page layout until needed

Designing with Layers: Analyst Considerations

- **How is the layer built?** Predefined HTML/CSS? Generated by JavaScript? Data needed from server (using Ajax)?
- **What events create the layer?** Click, focus, key presses threshold?
- **What events remove the layer?** Click item, click outside, close button?
- **Where should the layer appear?** Specific x, y coordinates?
- **How large should the layer be?** Pixels? Handle overflow with scrollbars or paging?
- **What formatting?** Color, border? Hover effects for links?
- **What happens on selection?** Populate form fields? Navigate?

Demo: Update Insured Client - Residential vs. Update Insured Client - Corporate vs. Update Insured Client - Multiple Properties

Dynamic Web Pages: Pros, Cons, and Considerations

- **When to Use:** Good when parts of a form logically change based on other selections within the same overall task/form
- **When to Avoid:** If parts of the form need different security levels
- **Disadvantages**
 - **Requires JavaScript**
 - Won't work if users disable it
 - **Solution:** Provide clear message and instructions or have non-JavaScript alternative
 - **Accessibility**
 - Dynamic changes can be confusing for screen readers or assistive tech
 - Needs careful design and testing
- **Advantages:** Faster updates, fewer page loads, smoother UX

Improving Responsiveness: Ajax

- **What:** **A**synchronus **J**avaScript and **X**ML → Though XML is rarely used now, JSON is common
- **Goal:** Make web pages feel more like desktop applications → Faster, no constant reloading
- **Traditional Way:** User action → Request to server → Server processing → New page sent back
- **Ajax Way:** User action → Request to server via JavaScript in background → Server processing → Only needed data sent back (small text/JSON/XML) → JavaScript updates part of the current page without full reload
- **Result:** Smoother, faster, less waiting

Ajax: Design Choices and Trade-offs

- When to Use

- When small amounts of data are needed to update parts of a form/page
- Form/page logically belongs together
- Improves user experience for interactive tasks

- When to Avoid

- Clear separation of steps
- Major context shifts
- Complex security transitions between steps

Using Color in Display Design

- Purpose
 - Contrast foreground/background
 - Highlight important fields/data
 - Indicate errors
 - Call attention to special codes or status
- Key Principles
 - **High Contrast:** Essential for readability
 - **Limit Colors:** Use sparingly $\rightarrow \leq 4$ for novice users, ≤ 7 for experienced users
 - **Cultural Norms:** Be mindful of color meanings
 - **Accessibility:** Don't rely *only* on color $\rightarrow$ Use text/icons/position

Color Combinations and Accessibility

- **Legibility Matters:** Some combinations are easier to read than others
 - Black on Yellow
 - Green on White
 - Blue on White
 - White on Blue
 - Yellow on Black
- **Avoid:** Red on Green, Blue on Red → Low contrast, problematic for Color Vision Deficiency (CVD)
- **General Rule**
 - Brighter colors for foreground, less bright/muted for background
 - Assign contrasting colors first to elements that must be differentiated
- **Accessibility Reminder (CVD)**
 - Affects ~8-10% of males, < 1% females
 - Design must work even if colors are not perceived correctly
 - Use color as a redundant cue, not the only cue

Website Design: Common Elements and Conventions

- Goal
 - Professional look
 - Effective communication
 - Usability
- **Focus:** Common navigational and feedback elements
 - Hamburger Menu
 - Context-Sensitive Help
 - Input Validation
 - Breadcrumb Trail
 - Fat Footer

The Hamburger Menu

- **Appearance:** Three short, stacked horizontal bars
- **Placement:** Usually top-left or top-right corner of the display
- **Function:** Click/tap reveals a navigation menu
- **Benefit**
 - Saves valuable screen space
 - Especially critical on mobile devices
 - Keeps the main view uncluttered

Context-Sensitive Help

- What
 - Help tied to the specific item the user is using
 - Shows tips *only* for that element
- How
 - Cursor turns into a ? over certain elements
 - Clicking shows a tooltip or pop-up → E.g., “Password must be at least 8 characters”
- Benefit
 - No need to search external help pages
 - Keeps users focused and reduces frustration

Input Validation

- **Recap:** Prevents entry of incorrect data types or formats before submission
- **How It Works**
 - Prevents user from moving to the next field until the current one is valid
 - Provides immediate, clear error messages near the field
- **Benefits**
 - Real-time error correction by the user
 - Ensures higher data quality before it hits the server/database

Breadcrumb Trail Navigation

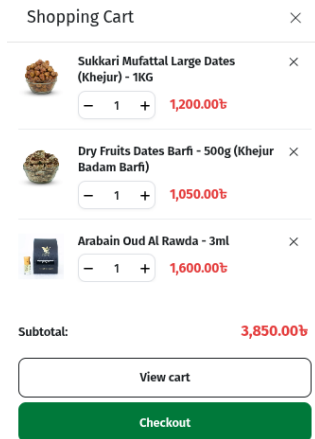
- **What:** Shows the user's current location within the website hierarchy
- **Analogy:** Dropping breadcrumbs to find your way back
- **Appearance:** Series of links showing the path taken
 - Usually at the top of the page below the main header
 - **Example:** Home > Products > Electronics > Laptops
- **Benefit**
 - Helps users understand where they are
 - Easily navigate back up levels
 - Enhances sense of control on large sites

The Fat Footer

- **What:** An expanded footer area used strategically
- **Potential Content**
 - Key site links → About Us, FAQ, Site Map, Careers
 - Social media links
 - Contact information → Address, Phone
 - Links to popular sections
 - SEO keywords/tags
 - Newsletter signup
 - Mission statement excerpt, slogan, values
 - Trust seals, awards
- **Purpose**
 - Go beyond basics to provide value
 - Aid navigation
 - Improve SEO
 - Enhance branding

E-commerce Specifics: Building Confidence

- **Challenge:** Users need confidence when buying online
- **Key Areas**
 - Buying the correct item and quantity
 - Getting the right price
 - Understanding the total cost (including shipping, taxes)
- **Common Solution:** The Shopping Cart/Shopping Bag metaphor
 - Allows users to review selections before committing
 - Provides control to modify order → Change quantity, remove items



Broader Ecommerce Design Objectives

- Design should reflect business goals and values
- **Key Objectives**
 - Communicate corporate mission/values
 - Ensure efficient and secure transaction processing
 - Build positive customer relationships
 - Preserve user privacy
- **Analyst Role:** Design must support these broader business and user needs

References I

- [1] J. Nielsen and A. Li, “Mega menus work well for site navigation,” (2017), [Online]. Available: <https://www.nngroup.com/articles/mega-menus-work-well/> (cit. on pp. 49–52).